

Section 5.4

I. First Fundamental Theorem of Calculus *If f is continuous on the interval $[a, b]$ and if*

F is an anti-derivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$

II. Examples

A. Definition

B. Fundamental Theorem of Calculus

III. Area under a curve

IV. Second Fundamental Theorem of Calculus *Let f be continuous on the interval $[a, b]$.*

Define the function G by $G(x) = \int_a^x f(t) dt$. Then, G is differentiable on $[a, b]$ and

$G'(x) = f(x)$, i.e., G is an anti-derivative of f .

V. Examples

VI. Extension